NUCLEAR ISOTOPE IDENTIFICATION LAB

Evidence report | v1.0.0 | 2026-10-15T09:01:32.716Z

Input provenance: Declared simulated example; not an experimental spectrum

Data: IAEA-LiveChart-ENSDF-subset-2026-10-14

Overall status: Supported (uncalibrated classification)

Research evidence report; not a definitive isotope certificate

MEASUREMENT & ANALYSIS SETTINGS

Energy range: 20–3500 keV. Minimum absolute emission probability: 0.1%. Matching multiplier k: 3.

Calibration assessment supplied: true. Common calibration standard uncertainty: 0 keV. Known offset subtracted: 0 keV.

Fallback window components: 0.4 keV and 800 ppm. This window is not a statistical uncertainty.

Efficiency error correlation: 0. Summing treatment: negligible. Extra response-ratio relative uncertainty: 0.

Peak list declared complete: false. External detection threshold: unavailable net counts. Mixture maximum: 3 components. Centroid-model inlier fraction: 0.9.

DETECTOR FULL-ENERGY EFFICIENCY INPUTS

20 keV: absolute efficiency=0.1; u=0.005; summing factor=unavailable; u(summing)=unavailable.

500 keV: absolute efficiency=0.08; u=0.004; summing factor=unavailable; u(summing)=unavailable.

661.657 keV: absolute efficiency=0.07; u=0.0035; summing factor=unavailable; u(summing)=unavailable.

1,173.228 keV: absolute efficiency=0.04; u=0.002; summing factor=unavailable; u(summing)=unavailable.

1,332.492 keV: absolute efficiency=0.035; u=0.00175; summing factor=unavailable; u(summing)=unavailable.

1,460.82 keV: absolute efficiency=0.03; u=0.0015; summing factor=unavailable; u(summing)=unavailable.

3,500 keV: absolute efficiency=0.01; u=5.000e-4; summing factor=unavailable; u(summing)=unavailable.

OBSERVED PEAKS

P1: E=661.67 keV, u(E)=0.1, net area=5,957, u(area)=80, FWHM=1.6 keV, live time=600 s, detector=simulated-HPGe, external evidence weight=1.

P2: E=1,173.24 keV, u(E)=0.12, net area=3,994, u(area)=65, FWHM=2 keV, live time=600 s, detector=simulated-HPGe, external evidence weight=1.

P3: E=1,332.5 keV, u(E)=0.12, net area=3,499, u(area)=60, FWHM=2.1 keV, live time=600 s, detector=simulated-HPGe, external evidence weight=1.

P4: E=1,460.82 keV, u(E)=0.14, net area=2,132, u(area)=47, FWHM=2.2 keV, live time=600 s, detector=simulated-HPGe, external evidence weight=1.

P5: E=1,911.111 keV, u(E)=0.1, net area=120, u(area)=15, FWHM=2.4 keV, live time=600 s, detector=simulated-HPGe, external evidence weight=1.

CANDIDATE RANKING (EVIDENCE POINTS; NOT PROBABILITY)

Co-60: score=27.993, 2 independent supports, 2 exclusive. 10 x 2 independent support + 3 x 2 exclusive groups + 2 x 0.996 energy agreement. Missing-line contradictions are evaluated separately.

K-40: score=15, 1 independent supports, 1 exclusive. 10 x 1 independent support + 3 x 1 exclusive groups + 2 x 1.000 energy agreement. Missing-line contradictions are evaluated separately.

Cs-137: score=14.983, 1 independent supports, 1 exclusive. 10 x 1 independent support + 3 x 1 exclusive groups + 2 x 0.992 energy agreement. Missing-line contradictions are evaluated separately.

ALL ENERGY-COMPATIBLE REFERENCE LINES

P1 -> Cs-137 661.657 keV; delta=0.013 keV; z=0.13; window=0.300135 keV; basis=uncertainty.

Reference 137cs-L3: absolute P=0.851; u(P)=0.002; daughter=Ba-137; nuclear levels 661.659 -> 0 keV; mode=B-; parent level=0 keV.

Evaluation cutoff: 1-Oct-2006; evaluators: E. BROWNE and J. K. TULI; extraction: 2026-10-14; raw source: data/raw/137cs-gamma.csv, row 3.
https://www-nds.iaea.org/relnsd/v1/data?fields=decay_rads&nuclides=137cs&rad_types=g

P2 -> Co-60 1,173.228 keV; delta=0.012 keV; z=0.1; window=0.360112 keV; basis=uncertainty.

Reference 60co-L6: absolute P=0.9985; u(P)=3.000e-4; daughter=Ni-60; nuclear levels 2,505.748 -> 1,332.508 keV; mode=B-; parent level=0 keV.

Evaluation cutoff: 31-Dec-2012; evaluators: E. BROWNE and J. K. TULI; extraction: 2026-10-14; raw source: data/raw/60co-gamma.csv, row 6.
https://www-nds.iaea.org/relnsd/v1/data?fields=decay_rads&nuclides=60co&rad_types=g

P3 -> Co-60 1,332.492 keV; delta=0.008 keV; z=0.067; window=0.3602 keV; basis=uncertainty.

Reference 60co-L7: absolute P=0.999826; u(P)=6.000e-6; daughter=Ni-60; nuclear levels 1,332.508 -> 0 keV; mode=B-; parent level=0 keV.

Evaluation cutoff: 31-Dec-2012; evaluators: E. BROWNE and J. K. TULI; extraction: 2026-10-14; raw source: data/raw/60co-gamma.csv, row 7.
https://www-nds.iaea.org/relnsd/v1/data?fields=decay_rads&nuclides=60co&rad_types=g

P4 -> K-40 1,460.82 keV; delta=0 keV; z=0; window=0.420268 keV; basis=uncertainty.

Reference 40k-L2: absolute P=0.1066; u(P)=0.0013; daughter=Ar-40; nuclear levels 1,460.851 -> 0 keV; mode=EC+B+; parent level=0 keV.

Evaluation cutoff: 30-Sep-2015; evaluators: JUN CHEN; extraction: 2026-10-14; raw source: data/raw/40k-gamma.csv, row 2.
https://www-nds.iaea.org/relnsd/v1/data?fields=decay_rads&nuclides=40k&rad_types=g

UNEXPLAINED PEAKS

P5 (1,911.111 keV)

MULTI-LINE CONSISTENCY: Co-60

Independent lines=2; centroid chi-square=0.01443327; degrees of freedom=2; reduced chi-square=0.00721663; common calibration u=0 keV.

Gaussian centroid consistency with a shared calibration covariance; no calibrated identification probability.

P2: reference 1,173.228 keV; delta 0.012 keV; z=0.099969.

P3: reference 1,332.492 keV; delta 0.008 keV; z=0.06663.

LINE VISIBILITY: Co-60

1,173.228 keV (60co-L6): observed; expected mean=unavailable +/- unavailable, lower mean estimate=unavailable counts. Energy-compatible observed peak; shared assignments are not proof of this nuclide.

1,332.492 keV (60co-L7): observed; expected mean=unavailable +/- unavailable, lower mean estimate=unavailable counts. Energy-compatible observed peak; shared assignments are not proof of this nuclide.

EFFICIENCY-CORRECTED RATIOS: Co-60

1,332.492 / 1,173.228 keV: observed=0.876064 +/- 0.020711; expected=0.876162 +/- 0.061955; log residual=-1.117e-4; z=-0.001499. Compatible under the stated detector and summing model.

Emission-probability and summing-factor errors are treated as independent; their evaluation covariances are unavailable. Additional ratio-model uncertainty must be explicitly supplied for numerical residuals.

Ratios share the same anchor and are correlated. No independent-ratio chi-square is computed. Peak-area correlations are not supplied; overlapping peaks are excluded.

MULTI-LINE CONSISTENCY: K-40

Independent lines=1; centroid chi-square=0; degrees of freedom=1; reduced chi-square=0; common calibration u=0 keV.

Single independent line; multi-line consistency unavailable.

P4: reference 1,460.82 keV; delta 0 keV; z=0.

LINE VISIBILITY: K-40

1,460.82 keV (40k-L2): observed; expected mean=unavailable +/- unavailable, lower mean estimate=unavailable counts. Energy-compatible observed peak; shared assignments are not proof of this nuclide.

EFFICIENCY-CORRECTED RATIOS: K-40

Not assessable with supplied data.

At least two isolated, non-shared peaks with net areas, absolute emission probabilities, and full-energy efficiency coverage are required. No raw-area vs emission-probability comparison is made.

MULTI-LINE CONSISTENCY: Cs-137

Independent lines=1; centroid chi-square=0.0168848; degrees of freedom=1; reduced chi-square=0.0168848; common calibration u=0 keV.

Single independent line; multi-line consistency unavailable.

P1: reference 661.657 keV; delta 0.013 keV; z=0.129942.

LINE VISIBILITY: Cs-137

661.657 keV (137cs-L3): observed; expected mean=unavailable +/- unavailable, lower mean estimate=unavailable counts. Energy-compatible observed peak; shared assignments are not proof of this nuclide.

EFFICIENCY-CORRECTED RATIOS: Cs-137

Not assessable with supplied data.

At least two isolated, non-shared peaks with net areas, absolute emission probabilities, and full-energy efficiency coverage are required. No raw-area vs emission-probability comparison is made.

SHARED-LINE AMBIGUITIES

No shared-parent evidence groups under the selected library and filter.

DISCRIMINATING LINES

Suggested discriminating lines are separated under the declared fallback window only. Detectability and detector resolution must be checked; the database is incomplete.

Co-60: No unobserved separated line in this subset.

K-40: No unobserved separated line in this subset.

Cs-137: No unobserved separated line in this subset.

MIXTURE EXPLANATIONS

Maximum externally weighted group coverage minus 0.35 per component; top 12 candidates, at most 4 components

Searched subsets=7; candidate space truncated=false.

Alternative 1: Co-60 + K-40 + Cs-137; objective=2.95; independent-group coverage=80%; explained peaks=P1, P2, P3, P4; unexplained=P5.

P1: Cs-137

P2: Co-60

P3: Co-60

P4: K-40

P5: unexplained

Alternative 2: Co-60 + Cs-137; objective=2.3; independent-group coverage=60%; explained peaks=P1, P2, P3; unexplained=P4, P5.

P1: Cs-137

P2: Co-60

P3: Co-60

P4: unexplained

P5: unexplained

Alternative 3: Co-60 + K-40; objective=2.3; independent-group coverage=60%; explained peaks=P2, P3, P4; unexplained=P1, P5.

P1: unexplained

P2: Co-60

P3: Co-60

P4: K-40

P5: unexplained

Alternative 4: Co-60; objective=1.65; independent-group coverage=40%; explained peaks=P2, P3; unexplained=P1, P4, P5.

P1: unexplained

P2: Co-60

P3: Co-60

P4: unexplained

P5: unexplained

Alternative 5: K-40 + Cs-137; objective=1.3; independent-group coverage=40%; explained peaks=P1, P4; unexplained=P2, P3, P5.

P1: Cs-137

P2: unexplained

P3: unexplained

P4: K-40

P5: unexplained

Alternative 6: Cs-137; objective=0.65; independent-group coverage=20%; explained peaks=P1; unexplained=P2, P3, P4, P5.

P1: Cs-137

P2: unexplained

P3: unexplained

P4: unexplained

P5: unexplained

Alternative 7: K-40; objective=0.65; independent-group coverage=20%; explained peaks=P4; unexplained=P1, P2, P3, P5.

P1: unexplained

P2: unexplained

P3: unexplained

P4: K-40

P5: unexplained

EVIDENCE COVERAGE

Listed peaks explained: 4/5 (80%). Unexplained listed-peak fraction: 20%.

Independent groups explained: 4/5 (80%). Externally weighted evidence coverage: 80%.

Fraction of listed net peak area: 99.23577%.

Coverage is not an identification probability. Listed net peak areas exclude continuum and all unlisted counts; they do not measure the fraction of total spectrum counts.

ALTERNATIVE HYPOTHESES

Equal reference-line Gaussian centroid mixture with fixed inlier fraction and uniform clutter on the declared range; common calibration nuisance integrated. Relative likelihoods are not posterior probabilities.

Co-60 + K-40 + Cs-137: reference centers=4; log L=-11.60798083; Delta deviance=0; relative likelihood=1. Preferred under the stated conditional model. Conditional centroid density; line-selection, database coverage, and fixed clutter assumptions limit interpretation.

Co-60 + Cs-137: reference centers=3; log L=-20.75726158; Delta deviance=18.2985615; relative likelihood=1.063e-4. Less supported under the stated conditional model. Conditional centroid density; line-selection, database coverage, and fixed clutter assumptions limit interpretation.

Co-60 + K-40: reference centers=3; log L=-21.08546956; Delta deviance=18.95497745; relative likelihood=7.656e-5. Less supported under the stated conditional model. Conditional centroid density; line-selection, database coverage, and fixed clutter assumptions limit interpretation.

K-40 + Cs-137: reference centers=2; log L=-30.54780348; Delta deviance=37.87964531; relative likelihood=5.950e-9. Less supported under the stated conditional model. Conditional centroid density; line-selection, database coverage, and fixed clutter assumptions limit interpretation.

Co-60: reference centers=2; log L=-30.5745291; Delta deviance=37.93309653; relative likelihood=5.793e-9. Less supported under the stated conditional model. Conditional centroid density; line-selection, database coverage, and fixed clutter assumptions limit interpretation.

Cs-137: reference centers=1; log L=-40.56008874; Delta deviance=57.90421583; relative likelihood=2.668e-13. Less supported under the stated conditional model. Conditional centroid density; line-selection, database coverage, and fixed clutter assumptions limit interpretation.

K-40: reference centers=1; log L=-40.88830299; Delta deviance=58.56064433; relative likelihood=1.922e-13. Less supported under the stated conditional model. Conditional centroid density; line-selection, database coverage, and fixed clutter assumptions limit interpretation.

NUCLIDE-SPECIFIC CONCLUSIONS

Co-60: Supported. 2 independent support(s), 2 exclusive group(s), 0 conditional contradiction(s).

K-40: Tentative. 1 independent support(s), 1 exclusive group(s), 0 conditional contradiction(s). One line is insufficient for a multi-line identification.

Cs-137: Tentative. 1 independent support(s), 1 exclusive group(s), 0 conditional contradiction(s). One line is insufficient for a multi-line identification.

LIMITATIONS & WARNINGS

This application provides research-oriented gamma-ray isotope identification support. Candidate identifications are not definitive without suitable measurement quality, calibrated detector response, validated analysis methods, and expert review.

Evidence scores and verbal classifications are uncalibrated research rules, not probabilities or statistical confidence levels.

This is a finite, filtered 11-parent-state library. Other nuclides, artefacts, escape peaks, sum peaks, and background can explain observations.

No activity fractions or nuclide concentrations are inferred.

The peak list has not been declared complete above a known threshold. Unobserved lines are not negative evidence.

MODEL VALIDATION

Independent experimental campaign: false. Calibrated evidence classes: false.

REFERENCE DATA MANIFEST

Data version: IAEA-LiveChart-ENSDF-subset-2026-10-14; stored lines=667; parent states=Co-60, Cs-137, Na-22, Eu-152, Ba-133, Am-241, I-131, Tc-99m, K-40, Bi-214, Pb-214.

API guide describes an April 2022 ENSDF snapshot. Per-line evaluation publication cutoff and actual extraction date are retained; extraction date is NOT an evaluation date. No assertion of latest ENSDF.

Co-60: included lines=6; source records=20; parent selection level=0 keV; raw file=data/raw/60co-gamma.csv.

Source SHA-256: 6e7296eb9661e9bd75034bb1382749ddcc8a900662b9e7c6e8d088baf2d2e626

https://www-nds.iaea.org/relnsd/v1/data?fields=decay_rads&nuclides=60co&rad_types=g

Cs-137: included lines=2; source records=8; parent selection level=0 keV; raw file=data/raw/137cs-gamma.csv.

Source SHA-256: 838cb4f251a6eb21a69738deb8a74dd3ca9a401ffb5f949e25ce0f89494a9633

https://www-nds.iaea.org/relnsd/v1/data?fields=decay_rads&nuclides=137cs&rad_types=g

Na-22: included lines=1; source records=4; parent selection level=0 keV; raw file=data/raw/22na-gamma.csv.

Source SHA-256: a95211d5310668f81df3b1494138f9454592935b3f5252510a66ae606daa9a4a

https://www-nds.iaea.org/relnsd/v1/data?fields=decay_rads&nuclides=22na&rad_types=g

Eu-152: included lines=173; source records=308; parent selection level=0 keV; raw file=data/raw/152eu-gamma.csv.

Source SHA-256: 247527b389b03484bca8833ffa24b77113ef42ce378a199939504e7eaa75dd9c

https://www-nds.iaea.org/relnsd/v1/data?fields=decay_rads&nuclides=152eu&rad_types=g

Ba-133: included lines=9; source records=31; parent selection level=0 keV; raw file=data/raw/133ba-gamma.csv.

Source SHA-256: 5b493bb07f9efce0c017610b216469d2fcdbc8a8bdbbefcd6205c2809f2c14dd

https://www-nds.iaea.org/relnsd/v1/data?fields=decay_rads&nuclides=133ba&rad_types=g

Am-241: included lines=128; source records=179; parent selection level=0 keV; raw file=data/raw/241am-gamma.csv.

Source SHA-256: f1f0d205ee2959980899fa11f70516a3d8ec19f4154db16b4d59d9954314fdc1

https://www-nds.iaea.org/relnsd/v1/data?fields=decay_rads&nuclides=241am&rad_types=g

I-131: included lines=19; source records=26; parent selection level=0 keV; raw file=data/raw/131i-gamma.csv.

Source SHA-256: b714da1525de09511133dd223b22255dffa76aeed5d12ca34f2068c238909f97

https://www-nds.iaea.org/relnsd/v1/data?fields=decay_rads&nuclides=131i&rad_types=g

Tc-99m: included lines=5; source records=25; parent selection level=142.6835 keV; raw file=data/raw/99tc-gamma.csv.

Source SHA-256: 2f476cea3a69c71fc76b1dad7c7d213a1f3ae67c5dab2aed7417262bc903d9bd

https://www-nds.iaea.org/relnsd/v1/data?fields=decay_rads&nuclides=99tc&rad_types=g

K-40: included lines=1; source records=7; parent selection level=0 keV; raw file=data/raw/40k-gamma.csv.

Source SHA-256: c23bd09f5ff1094ea93a714878cdbd73cc7fc883ab9205af082f2a131f4caaf9

https://www-nds.iaea.org/relnsd/v1/data?fields=decay_rads&nuclides=40k&rad_types=g

Bi-214: included lines=302; source records=322; parent selection level=0 keV; raw file=data/raw/214bi-gamma.csv.

Source SHA-256: aeb2c4d2f998208f72d0a99a793a4243b587e7e3942fe951983c0f7ae31e3f54

https://www-nds.iaea.org/relnsd/v1/data?fields=decay_rads&nuclides=214bi&rad_types=g

Pb-214: included lines=21; source records=35; parent selection level=0 keV; raw file=data/raw/214pb-gamma.csv.

Source SHA-256: dd9d4d7d9c817b4d171e9dfdccddd44bc2d50842f47dc59ebe50b9a6e9fe0861

https://www-nds.iaea.org/relnsd/v1/data?fields=decay_rads&nuclides=214pb&rad_types=g